

Where do Stories Live

A Study of Estonian Runosongs' Themes

Network Science Project: Liisa Jullinen

Aim, data

The aim of this study was to analyze the spread of Estonian runosongs' themes along geographical locations to understand possible patterns of cultural transmission. It was also of interest which thematic clusters would form and which songs contribute most to cluster forming, e.g. which songs are more universal and which more place-specific.

A **dataset** from the Estonian Runosongs' Database [1] (by Estonian Folklore Archives) was retrieved and after data cleaning 70352 song records from 2703 song types and from 104 different Estonian parishes (est kihelkond) were analyzed.

Methods

Lists of songs recorded in each parish were formed, parishes were treated as nodes and shared song types were treated as edges (number of shared songs as weights) and a graph created (Fig 2). The **Louvain method** was used to **detect non-overlapping communities** of the runosongs. The resulting (fully connected) **graph was mapped** over the Estonian map of parishes (Fig 1) [2].

Since separate songs can be viewed as "layers" of the bigger graph, **songs that contributed more to cluster formation were specified** and some of the individual song graphs specific to clusters were also illustrated (Fig 3).

Node2vec and K-means clustering was also conducted, without an improvement in results (thus not depicted here).

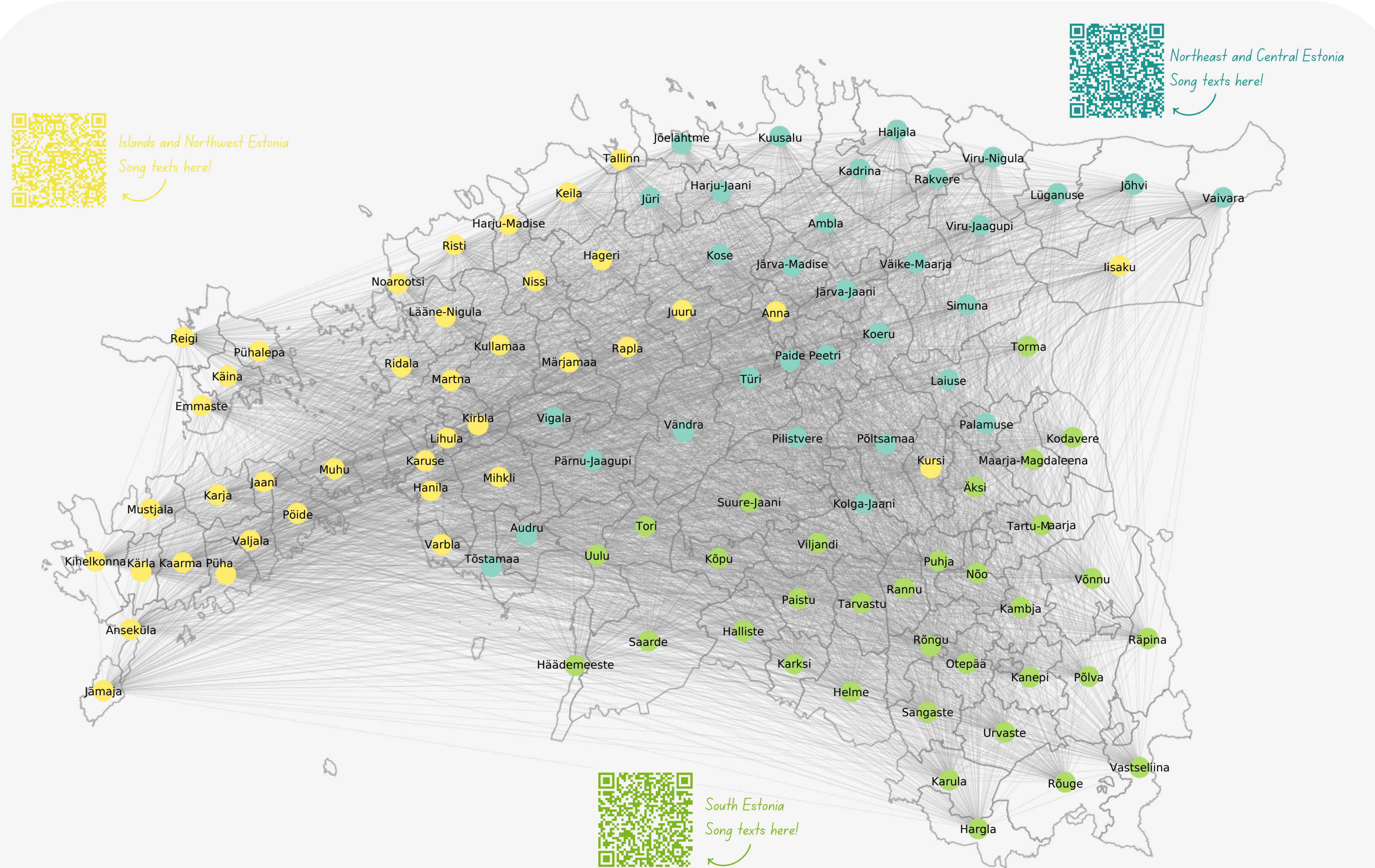


Figure 1: Clustered network mapped on the geographical map of Estonia (1917)
To view most influential songs for cluster formation, and song texts, scan the QR codes.

Results

The Louvain method **detected 3 communities**, that have rather **clear geographical locations** (southern Estonia, Islands and Northwest of Estonia and Northeastern and central part of the country). These areas are in accordance very well both with dialect borders of runosongs (the southern Estonian dialect border), historical borders (e.g. Polish/Swedish rule) and also the theoretical framework of the Estonian folklore areas (although this analysis did not take these factors into account).

Also **the most influential songs for community formation** were found (Fig 1).

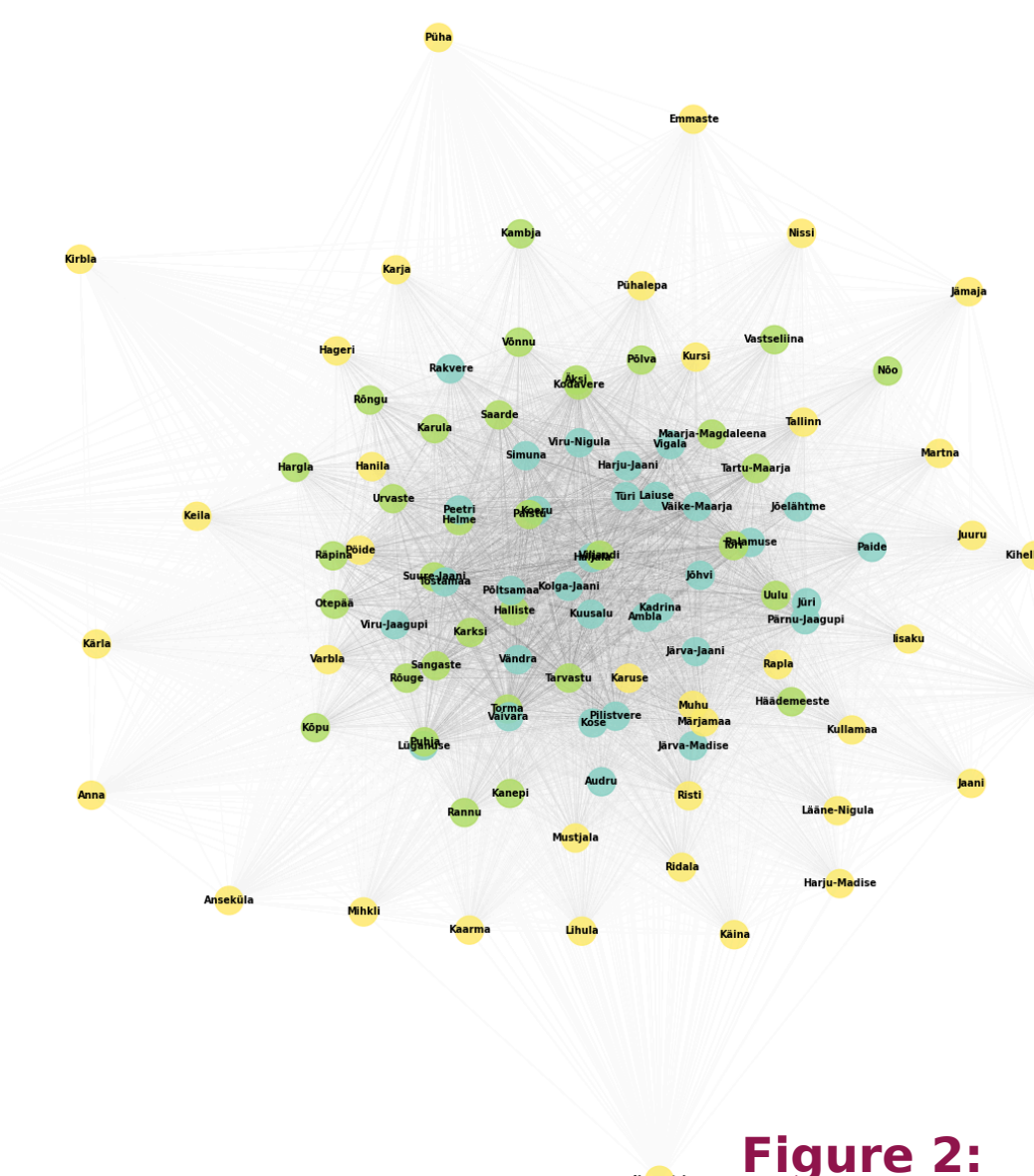


Figure 2: Clustered network

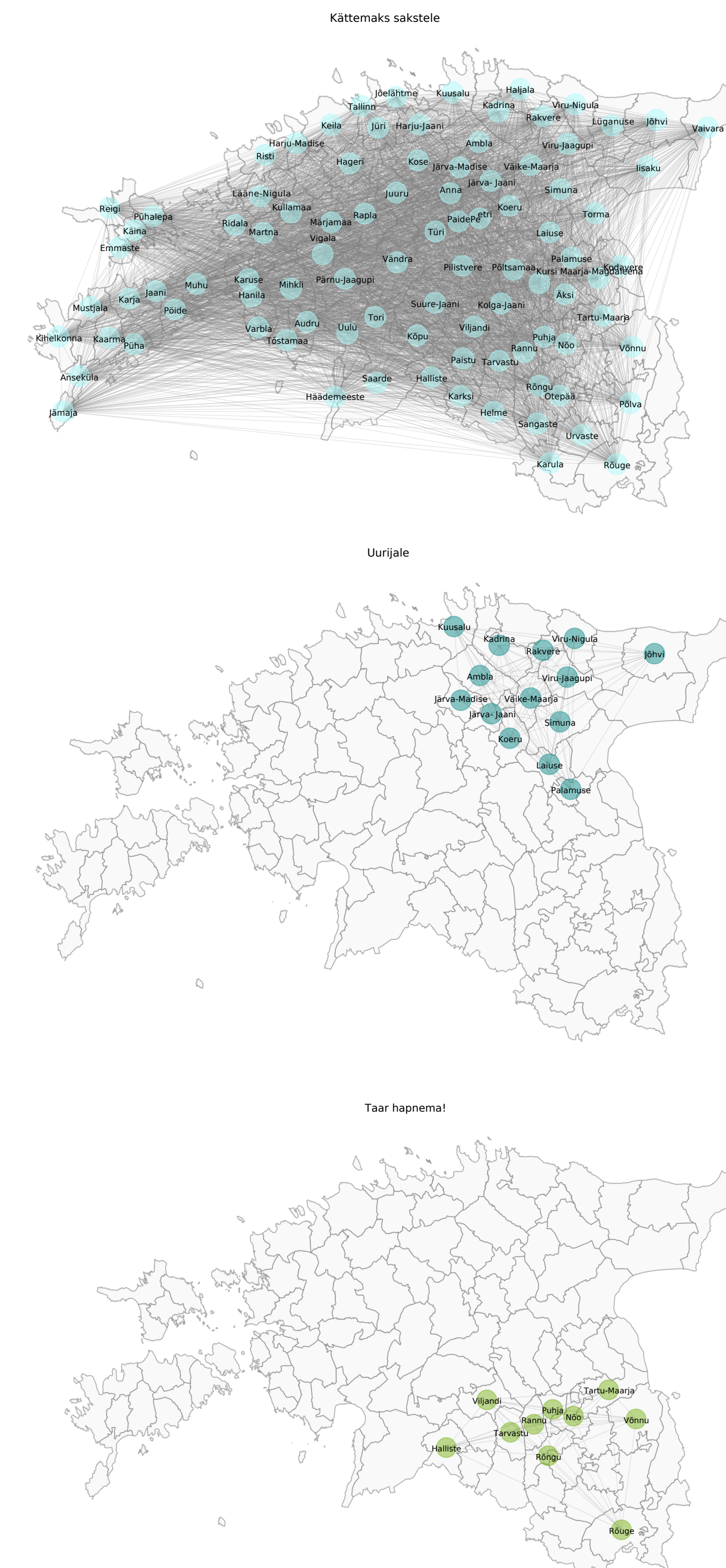


Figure 3: Song networks
As visible above, some songs are more universal and others contribute more to cluster formation

[1] <https://www.folklore.ee/regilaul/andmebaas/>
[2] https://geoportaal.maaamet.ee/docs/haldus_asustus/ajaloolised/Kihelkond_kuni_1917.zip